The Observable Dynamics Program: A Categorical Reading

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Abstract

This note develops the Observable Dynamics Program in categorical language, as a direct companion to the classical picture in *categorical_classical.tex*. The classical construction is algebra-first: a probability measure is freely chosen and marginals are derived from it. The program is diagram-first: the measure is derived from the coherence of a system of marginals. We trace this inversion through the same categorical layers — **Meas**, the Giry monad $\mathcal{P}$, its Kleisli category $\mathrm{Kl}(\mathcal{P})$ of Markov kernels, and the extension theorem — and identify where the program's arguments must go and what remains open. The inversion amounts to asking whether the forgetful functor from probability spaces to compatible marginal systems has a left adjoint, and under what conditions.

Contents

1 The inversion: diagram-first probability

The classical construction of probability (see *categorical_classical.tex*, §6) has a single free parameter: the choice of measure $P \in \mathcal{P}X$ at Step 3. Everything else — pushforwards, marginals, kernels, projective limits — is forced by the categorical structure of **Meas** and $\mathrm{Kl}(\mathcal{P})$.

The Observable Dynamics Program asks: can Step 3 be replaced by a derivation? Specifically: given a projective system of measurable spaces and a compatible family of probability measures on its finite-dimensional factors, is there a *canonical* probability measure on the projective limit that is forced by the coherence data, rather than freely chosen?

The program's answer is yes — under conditions that we now identify categorically.

Classical (algebra-first)	Program (diagram-first)
Choose $P \in \mathcal{P}X$ freely	Derive P from cocone data
Marginals follow from P	P follows from marginals
Monad algebra assumed	Monad algebra forced
Step 3 is a free parameter	Step 3 is a theorem

2 Query systems as diagrams in $\mathrm{Kl}(\mathcal{P})$

Definition 1 (Query system)**.** *A* query system $(\mathcal{Q}, \preceq)$ *consists of:*

- *A directed preorder* $(I, \leq)$ *of* query indices.
- *For each* $i \in I$*, a measurable space* $(O_i, \mathcal{O}_i)$ *— the* outcome space *of query* i.
- *For each* $i \leq j$*, a measurable function* $\pi_{ij} : O_j \to O_i$ *— the* refinement map *from finer query* j *to coarser query* i *— satisfying* $\pi_{ii} = \mathrm{id}$ *and* $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$ *for* $i \leq j \leq k$.

Remark 1 (Query systems as functors)**.** *A query system is a functor* $F : I^{\mathrm{op}} \to \mathbf{Meas}$*, where* I *is the preorder category with objects the query indices and morphisms the order relations. This is precisely a* projective system *in* **Meas**.

The refinement maps π_{ij} *are deterministic, so they embed into* $\mathrm{Kl}(\mathcal{P})$ *via the Dirac kernel* $\hat{\pi}_{ij}(o, \cdot) = \delta_{\pi_{ij}(o)}$*. The query system therefore lives in* **Meas***, but it is a diagram in* $\mathrm{Kl}(\mathcal{P})$ *under this embedding. We work in* $\mathrm{Kl}(\mathcal{P})$ *because the morphisms we ultimately care about — compatible marginals and extension kernels — are genuinely stochastic.*

Definition 2 (Realization space)**.** *The* realization space *of a query system is the projective limit*

$$\Omega = \varprojlim_{i \in I} O_i = \left\{ (o_i)_{i \in I} \in \prod_i O_i \;\middle|\; \pi_{ij}(o_j) = o_i \text{ for all } i \leq j \right\}$$

equipped with the subspace σ-algebra from the product. The projections $\mathrm{pr}_i : \Omega \to O_i$ *are the canonical measurable maps.*

Remark 2 (What Ω represents)**.** *A point* $\omega \in \Omega$ *is a* coherent tuple *of query outcomes: a choice of answer to every query simultaneously, consistent across all refinement relations. This is the program's analogue of a point in the sample space* X *— except* Ω *is constructed from the queries, not assumed.*

In the delay query system (Paper 4 / observational_probability.tex*),* $\Omega \subseteq X^{\mathbb{Z}}$ *is the set of bi-infinite sensor streams that are coherent under all delay-and-sample operations.*

3 Compatible marginals as cocones

Definition 3 (Compatible marginals)**.** *A* compatible family of probability measures *for a query system* $(\{O_i\}, \{\pi_{ij}\})$ *is a collection* $\{\nu_i \in \mathcal{P}O_i\}_{i \in I}$ *satisfying*

$$(\pi_{ij})_\sharp \nu_j = \nu_i \qquad \text{for all } i \leq j.$$

Remark 3 (Cocone interpretation)**.** *In* $\mathrm{Kl}(\mathcal{P})$*, the compatible family* $\{\nu_i\}$ *is a* cocone *over the diagram* $F : I^{\mathrm{op}} \to \mathrm{Kl}(\mathcal{P})$*. Specifically, each* ν_i *is a morphism* $1 \to O_i$ *in* $\mathrm{Kl}(\mathcal{P})$ *(a stochastic map from the terminal object to* O_i*), and the compatibility condition says that these morphisms commute with the refinement maps:*

$$\begin{tikzcd} 1 \arrow[r, "\nu_j"] \arrow[dr, "\nu_i"'] & O_j \arrow[d, "\hat{\pi}_{ij}"] \\ & O_i \end{tikzcd}$$

The apex of the cocone is (if it exists) a measure P on Ω with $(\mathrm{pr}_i)_\sharp P = \nu_i$ for all i.

The program's central question is: does this cocone have a canonical apex?

Remark 4 (Classical vs. program direction)**.** *Classically, the apex P is freely chosen first, and the cocone data $\{\nu_i\}$ are derived as pushforwards $(\mathrm{pr}_i)_\sharp P$. This is a* cone*: P is the apex and the ν_i are its projections.*

The program runs in the opposite direction: the cocone data $\{\nu_i\}$ are given (from observations), and the apex P is derived. This is the categorical form of the epistemological inversion: the global measure is constituted by the coherence of its parts, not given as a prior that determines the parts.

4 The extension theorem as an adjunction

We now state the program's main theorem — the Observational Extension Theorem of Paper 1 — in categorical language.

Definition 4 (Categories for the adjunction)**.** *Define two categories:*

- $\mathbf{Prob}(\Omega)$*: the category whose objects are probability measures $P \in \mathcal{P}\Omega$ and whose morphisms are measurable maps of probability spaces over Ω. (For route-finding purposes, we may take this to be a discrete category: objects are measures, no non-trivial morphisms.)*
- $\mathrm{Compat}(\{O_i\})$*: the category of compatible families $\{\nu_i\}$ over the query system, with morphisms being families of measure-preserving maps compatible with the refinements.*

Definition 5 (The forgetful functor)**.** *The* marginalisation functor

$$\Phi : \mathbf{Prob}(\Omega) \to \mathrm{Compat}(\{O_i\})$$

sends a probability measure P on Ω to the compatible family of its marginals:

$$\Phi(P) = \{(\mathrm{pr}_i)_\sharp P\}_{i \in I}\,.$$

This functor is always well-defined: pushforward is functorial and the compatibility condition holds automatically.

Theorem 1 (Extension theorem — adjunction form)**.** *Under the conditions of Paper 1 (query system sequentially upper-directed and realizable), the functor Φ is an* isomorphism of categories*: every compatible family $\{\nu_i\}$ extends uniquely to a measure P on Ω, and $\Phi(P) = \{\nu_i\}$.*

In particular, Φ has a two-sided inverse (the extension functor), so the adjunction $\mathrm{Ext} \dashv \Phi$ *is in fact an equivalence.*

Proof (sketch). The uniqueness direction — that Φ is injective — is the Observational Determination Theorem (Paper 1, Theorem 5.2): two measures on Ω with the same marginals on all query outcomes are equal, since the query σ-algebra generates $\mathcal{B}_\Omega$.

The existence direction — that Φ has a left inverse — is the Observational Extension Theorem (Paper 1, Theorem 6.1): the Carathéodory extension of the finitely additive content defined by the $\{\nu_i\}$ on cylinder sets is σ-additive, hence extends to a measure P on Ω. The key steps are:

1. Define μ_0 on cylinder sets via $\mu_0(\mathrm{pr}_i^{-1}(A)) = \nu_i(A)$.
2. Show μ_0 is a well-defined finitely additive content (using compatibility and upper-directedness of the query system).

3. Show μ_0 is σ-subadditive (using realizability to pass from empty cylinder intersections to empty intersections in Ω).

4. Apply the Carathéodory extension theorem to obtain P.

The Lean 4 formalization in `QuerySystem.lean` proves this in full generality; see `observational_extension` (line 1450ff). □

Remark 5 (The Polish replacement)**.** *The classical Kolmogorov extension theorem obtains existence by a topological route: Polish spaces are regular enough that Prokhorov's tightness theorem guarantees inner regularity, which in turn forces σ-additivity.*

The program obtains existence by a purely measure-theoretic route: the realizability condition replaces inner regularity. Realizability says that the projections $\mathrm{pr}_i : \Omega \to O_i$ are surjective, so that the projective limit is "full enough" that vanishing marginals imply vanishing cylinder content.

This is why the two approaches are complementary rather than redundant: the program's route works in settings where the spaces need not be Polish (any standard measurable space with the right query structure), and the relationship to the Prokhorov/tightness route is precisely what the Conjecture 2 / CFA work on the `equicontinuity-cfa-conjecture` branch is probing.

5 The minimal predictive query in $\mathrm{Kl}(\mathcal{P})$

We now identify the objects of Paper 2 as morphisms in $\mathrm{Kl}(\mathcal{P})$.

Definition 6 (Delay query system)**.** *Fix a measurable sensor alphabet $(X, \mathcal{X})$ and work on the path space $X^{\mathbb{Z}}$ with the product σ-algebra. The* delay query system *has:*

- Queries*: pairs $(d, \tau) \in \mathbb{N}_{>0} \times \mathbb{N}_{>0}$ with outcome space $O_{d,\tau} = X^d$ and evaluation map*

$$\mathrm{ev}_{d,\tau} : X^{\mathbb{Z}} \to X^d, \qquad \omega \mapsto (\omega_0, \omega_{-\tau}, \omega_{-2\tau}, \ldots, \omega_{-(d-1)\tau}).$$

- Refinement maps*: $\pi_{(d,\tau),(d',\tau')} : X^{d'} \to X^d$ defined when $(d, \tau) \preceq (d', \tau')$, projecting to the corresponding coordinates.*

The realization space is $\Omega \subseteq X^{\mathbb{Z}}$, the set of streams coherent under all delay evaluations.

Definition 7 (Predictive kernel as Kleisli morphism)**.** *Given a compatible family $\{\nu_{d,\tau}\}$ and the extended measure P on Ω (from Theorem 1), the* predictive kernel *at time t is a morphism in* $\mathrm{Kl}(\mathcal{P})$*:*

$$K_t : \Omega \to O_F, \qquad K_t(\omega, \cdot) = P\big((\sigma^t \omega \in \cdot) \mid \mathcal{F}^0_{-\infty}\big),$$

where σ is the shift map and $O_F = X^{\mathbb{N}}$ is the future observable space. The minimal predictive query *Q_* is the σ-algebra generated by $\{K_t\}_{t \geq 0}$ — the coarsest measurable structure on Ω from which all future conditional distributions can be computed.*

Remark 6 (Semigroup in $\mathrm{Kl}(\mathcal{P})$)**.** *The Markov property of $\{K_t\}$ is exactly closure of this collection under Chapman–Kolmogorov composition in* $\mathrm{Kl}(\mathcal{P})$*:*

$$K_{s+t} = K_t \circ K_s \qquad \textit{(equality in } \mathrm{Kl}(\mathcal{P})\textit{)}.$$

The semigroup property of the predictive operator (Paper 2, Theorem 5.2) is associativity of composition in $\mathrm{Kl}(\mathcal{P})$*. The operator-theoretic content of Paper 3 (resolvent, spectral theory) is the analysis of this semigroup acting on $L^2(\Omega, P)$.*

Remark 7 (The Rose condition as isomorphism)**.** *Two delay embeddings (via different (d, τ) parameters) are* observationally equivalent *if their induced predictive sub-diagrams are isomorphic in* $\mathrm{Kl}(\mathcal{P})$*. The Rose condition — $D_f(\Gamma_Y \| \Gamma_Z) = 0$ — is the information-theoretic criterion for this isomorphism. It says that no f-divergence can distinguish the two predictive distributions, which in* $\mathrm{Kl}(\mathcal{P})$ *means the two kernels are equal as morphisms.*

6 Open problems as categorical questions

The program's open problems now have precise categorical formulations.

Conjecture 1 (Conjecture 2 — σ-additivity from coherence alone)**.** *There exist purely algebraic conditions on the query system — without realizability or Polish hypotheses — that force the finitely additive content μ_0 to be σ-additive.*

Categorically*: what conditions on a diagram $F : I^{\mathrm{op}} \to \mathrm{Kl}(\mathcal{P})$ guarantee that the colimit exists in $\mathcal{P}\Omega$? This is a cocompleteness question for* $\mathrm{Kl}(\mathcal{P})$ *restricted to diagrams of a particular shape.*

Connection to equicontinuity*: the CFA (Compact Family Argument) work on the* `equicontinuity-cfa-con` *branch is developing the Prokhorov-type tightness argument for this. If successful, it would give a topological route to cocompleteness analogous to the classical case, and potentially feed into a Mathlib formalization of the Kolmogorov extension theorem.*

Conjecture 2 (Stopping Point 1 — measurability from discriminability)**.** *The σ-algebra $\mathcal{X}$ on the sensor alphabet X should be derivable from primitive discriminative structure: the ability of the observer to distinguish finite families of events.*

Categorically*: is there a category more primitive than* **Meas** *— perhaps of* locales, domains, *or* topological spaces *— whose image in* **Meas** *under a forgetful functor gives the right σ-algebras? The σ-algebra would then be the* sheaf of reportable distinctions *on a suitable site, and the assumption of $(X, \mathcal{X})$ in the current program would be replaced by a construction.*

Connection to Boole*: Boolean algebras of propositions (Boole's algebra of thought) are the finite discriminative structure; the σ-algebra is their countable closure. The conjecture is that any system capable of coherent countable probabilistic reasoning must already have this closure.*

Remark 8 (The adjunction gap)**.** *Theorem 1 states the extension theorem as an isomorphism of categories. The full adjunction statement — identifying the unit and counit, showing they satisfy the triangle identities, and characterising the fixpoints — has not been written out precisely. This is a concrete mathematical task: define the two categories properly, write down the functors Φ and* Ext*, and verify the adjunction axioms. It would give the cleanest categorical statement of what Paper 1 proves.*

7 Comparison table: classical vs. program

Step	**Classical**	**Program**
1. Starting point	Sample space $(X, \mathcal{X})$ assumed	Query system $(\{O_i\}, \{\pi_{ij}\})$ given
2. Ambient category	**Meas** and $\mathrm{Kl}(\mathcal{P})$	**Meas** and $\mathrm{Kl}(\mathcal{P})$ (same)
3. Measure	Free choice $P : 1 \to \mathcal{P}X$	Derived: $P = \mathrm{Ext}(\{\nu_i\})$
4. Marginals	Derived from P	Given as input (observations)
5. Limit space	X is assumed	$\Omega = \varprojlim O_i$ is constructed
6. Extension	Kolmogorov (Polish)	Paper 1 (sequential, realizable)
7. Dynamics	Markov kernel $T : X \to X$ in $\mathrm{Kl}(\mathcal{P})$	Predictive kernel $K_t : \Omega \to O_F$
8. Semigroup	Chapman–Kolmogorov	Composition in $\mathrm{Kl}(\mathcal{P})$ (same)
9. Objectivity	Ground truth assumed	Limit of coherent query perspectives

Row 3 is the inversion. Rows 2 and 8 show that the categorical machinery is the same in both directions — the difference is not in the tools but in the direction of travel.

Row 9 deserves emphasis: the program does not eliminate objectivity but *derives* it. When all queries cohere — when the compatible family $\{\nu_i\}$ is globally consistent — the extension P is the unique probability measure on Ω that everyone with access to all queries would agree on. This is the mathematical form of the phenomenological claim that objectivity is constituted by the coherence of perspectives, not presupposed beneath them.

8 Mathlib availability for the program's picture

- **Available**: `CategoryTheory.Monad.Kleisli` gives the general Kleisli construction. Instantiated at the Giry monad, morphisms are Markov kernels (`Probability.Kernel`), which are extensively developed.

- **Available**: `CategoryTheory.Monad.Algebra` gives the Eilenberg–Moore category and the free/forgetful adjunction. The query system adjunction (Theorem 1) is a specific instance of this pattern.

- **Available**: `MeasureTheory.Measure.GiryMonad` for the monad laws; `MeasureTheory.Constructions.P` for finite products; `MeasureTheory.Constructions.Projective` for projective families (partial).

- **Built in-house**: `QuerySystem.lean` provides `observational_extension` — the main theorem — as new Lean 4 infrastructure not yet in Mathlib. This fills the Kolmogorov extension gap for the query system setting.

- **Gap**: the precise adjunction of Theorem 1 (defining $\mathbf{Prob}(\Omega)$ and $\mathrm{Compat}(\{O_i\})$ as Lean 4 categories and verifying the adjunction axioms) has not been formalized. This is the next categorical formalization target.